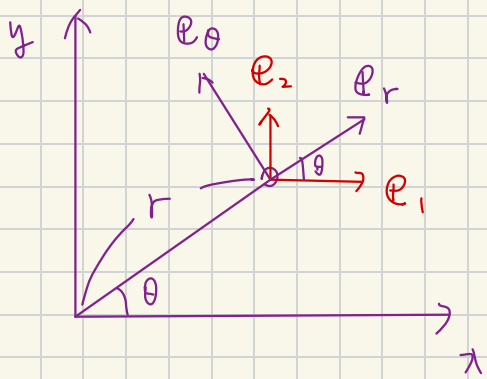
スライドにない部分

$\dot{e}_r$ $\dot{e}_\theta$ の具体形



方針

1. 「e_r, e_θ を 「e_1, e_2」で表現

2. t で微分

Proof

$$\begin{cases} e_r = \cos\theta e_1 + \sin\theta e_2 \\ e_\theta = -\sin\theta e_1 + \cos\theta e_2 \end{cases}$$

$$\dot{e}_r = (-\sin\theta)\frac{d\theta}{dt} e_1 + \cos\theta \cdot 0$$

$$+ \cos\theta \frac{d\theta}{dt} e_2 + \sin \cdot 0$$

$$= (-\sin\theta\, e_1 + \cos\theta\, e_2)\,\dot{\theta}$$

$$= \dot{\theta}\, e_\theta$$

$$\dot{e}_\theta = (-\cos\theta\, e_1 - \sin\theta\, e_2)\,\dot{\theta}$$

$$= -\dot{\theta}\, e_r$$

$$\boxed{\begin{aligned} \dot{e}_r &= \dot{\theta}\, e_\theta \\ \dot{e}_\theta &= -\dot{\theta}\, e_r \end{aligned}}$$